Supplementary Material for AnchorMTM

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Abstract

This document supplements the paper "AnchorMTM: Spatially Anchored Merge Tree Maps with Position Bounds." It gives the full definitions, proofs and implementation details for the bounds and AnchorMTM, per-frame distributions of AnchorMTM over the merge budget (Figs. S1–S4), the evaluation of the bounds on an earlier benchmark (Figs. S5 and S6), and an exploratory volumetric case.

S1 Overview

This companion gives the full definitions, proofs and implementation details for the bounds and AnchorMTM (Spatially Anchored Merge Tree Maps). Spatial Anchoring uses a fixed spatial projection as the target for display anchors, with deviations determined by topology and interval constraints. Statements use join convention; split trees are handled by negating the scalar field. Tree nodes with equal merge levels form multifurcations.

Correspondence provenance and temporal confidence. The shared fixed-grid extractor in `experiments/era5.py` calls `match_leaves_by_overlap` on consecutive augmented-tree frames. Supports contain the extremum and leaf-arc regular samples, excluding the parent critical sample. Stable `sample_ids` identify common spatial samples; fixed-grid C-order indexing supplies them in this extractor. A Hungarian assignment maximizes total intersection count, and zero-overlap pairs are discarded. Matched leaves inherit Track IDs; unmatched current leaves receive new IDs. The first frame and declared temporal-window breaks start identities without a predecessor. Tied assignments have no additional physical tie-breaking criterion.

The shared 2D/3D extraction path and the native-volume paths must be distinguished. The latter accumulate intersections from original-voxel leaf-arc ownership arrays, exclude roots, and retain positive-overlap Hungarian pairs. They implement the same count-based assignment principle through different support representations, rather than all calling one matching function. The automatic-reference adapter accepts precomputed, row-aligned Track IDs and checks frame-wise uniqueness; it does not verify their upstream matching provenance. External inputs must therefore declare their support and correspondence conventions.

For a persistent ID present in consecutive frames, `motion_pairs` supplies an extremum displacement to the single sequence-wide direction fit. Unmatched leaves and track interruptions supply no displacement for that step. Rapid motion, support changes or weak overlap may remove or alter correspondences and affect this fit. The implementation does not reconstruct physical split/merge events, branching lineages or links across missing frames.

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For placement, the adapter and `solve_unified_sequence` align previous anchors and references by Track ID and pass a valid-match mask. They use unit temporal confidence. Shared preprocessing also records support IoU for diagnostics; the legacy ERA5 extremum-reference branch additionally passes it as `temporal_confidence` to the low-level solver. That branch weights temporal residuals by IoU, but still maximizes intersection counts during matching. The current automatic-reference adapter does not enable this weighting; direction estimation also does not use IoU weights.

Automatic reference and implementation convention. The default preprocessing uses the leading eigenvector of the scatter of all matched extremum displacements, once over the full sequence in either 2D or 3D. It maximizes squared projected motion. The sign gives nonnegative total signed motion; when that sum is within 10^{-12} of zero, the largest-magnitude component is positive. If motion energy is at most 10^{-20}, the longest domain axis is used, choosing the first on a tie. These are the existing `directional_reference` conventions. A physical direction may be supplied. Repeated top eigenvalues need not select a unique direction; the relative eigengap is reported. This is standard spectral estimation, not joint optimization of direction and topology.

The projected bounding-box corners supply origin o and span L. Code stores raw $p_i^t \cdot a$ on $[o, o+L]$; the paper uses $q_i^t = p_i^t \cdot a - o$ and translates anchors and centers by the same origin. Motion differences are invariant under this shift. The low-level `solve_anchored_sequence` adapter connects this preprocessing to the finite width-aware search and four-term capped QP. It returns continuous layouts; the public `anchor_mtm` entry additionally invokes existing reconstruction and diagnostics, as specified below. Neither output is an unconditional raster certificate. Existing fixed-axis experimental drivers and stored evidence have not been rerun or relabelled as automatic-reference experiments.

Single-control configuration and execution. The public execution is `prepare_fields` (or `prepare_sequence`) followed by `anchor_mtm(prepared, epsilon)`. Preprocessing and its fitted reference are reused for all budgets. The fixed numerical allowance is $\varepsilon_0 = 10^{-7}$; every order stage uses the same absolute $\eta = (\varepsilon_0 + \varepsilon)\Delta_F$. Default $\varepsilon = 0$ retains this allowance and is not strict Exact Mode. Zero labelled error is the mathematical case $\eta = 0$, obtained with both allowances zero. Neither 20% nor 100% limits the interface.

The default configuration uses the existing scale rules: width scale $s = 0.5L/\text{measure}(D)$, gap $g = L/240$, position slack $\beta_0 = L/120$, $\rho = 0.5$, and weights $(\lambda_R, \lambda_M, \lambda_G, \lambda_C) = (4, 0.5, 1, 0.1)$. All are fixed for a prepared sequence. Output resolution defaults to 2048 samples; insufficient raster capacity is a reported failure. Advanced fixed configurations retain these quantities rather than deleting constraints. Reference-aligned motion and term means are the default. An independently supplied reference cap or conflicting motion control is rejected by the new placement interface.

The canonical ordering implementation retains the existing threshold/selective flattening, midpoint and block-insertion strategies (four initial refinement seeds plus the exact tree, at most eight refinement passes). A collector records generated initial and trial permutations, retaining strict-tree orders and the point candidate. The deduplicated finite pool is evaluated for δ^*, τ and NROE, filtered by η, then selected lexicographically by (τ, NROE); an exact

tie uses the permutation lexicographically for reproducibility. The QP receives a singleton order and the explicit cap $\tau+\beta_0$. The older general-purpose `Controls` interface can use an absolute position cap and compare multiple orders by QP objective; it is not the default AnchorMTM entry. Historical experiment drivers and their outputs are unchanged.

Reconstruction calls the existing leaf/LCA filling and optimal-barrier correction. Records include candidate totals and admitted counts, selected order, τ, δ^*, anchors, reference displacement, NROE, continuous motion residual, and separately rendered errors and failures. NMLEmean and NMLEmax use absolute error over the full fixed-label merge matrix (including the diagonal), divided by Δ_F; ME is the RMS reference-motion residual divided by L. Rendered motion is also measured at rounded anchors. A successful computation may record a rendered budget violation; it does not certify full-tree label coverage or repair the output. The renderer's strict-extremum clearance remains separate from the continuous optimum.

S2 Interval and Error Conventions

For a frame, T is the supplied merge tree, $\mathscr{F}$ its leaf labels, $\mathscr{D}(v)$ the descendant leaves of node v, and $h(v)$ its scalar level. The matrix V records source LCA levels and fixed extremum values on the diagonal. References lie in $[0,L]$; widths W_i are positive in the interval model, with zero widths used only for its non-strict point-model limit. The support S_i contains the extremum and its leaf-arc regular samples, excluding the parent. Its centroid is their unweighted mean, and relative geometry uses Euclidean centroid distances. The frame representation is $M_t=(\pi_t,x_t,\psi_t)$.

For a fixed order π, introduce interval centers z_i and anchor slack $H_i=\rho W_i/2$, with $0\le\rho\le 1$. The valid-layout set $\mathscr{L}(\pi)$ consists of pairs (x,z) satisfying

$$\begin{aligned} W_i/2 \le z_i \le L-W_i/2, \qquad & 0\le x_i\le L,\\ |x_i-z_i|\le H_i, \qquad & \\ z_{\pi(k+1)}-z_{\pi(k)} \ge (W_{\pi(k)}+W_{\pi(k+1)})/2+g, \qquad & \end{aligned} \tag{S1}$$

for every adjacent pair, with minimum gap $g\ge 0$. A necessary capacity condition is $\sum_i W_i+(n-1)g\le L$; infeasible widths are reported rather than silently rescaled. We suppress t in within-frame analysis.

Actual reference displacement is $e_i=|x_i-q_i|$, with $E_\infty=\max_i e_i/L$. An order has normalized reference-order error

$$\mathrm{NROE}(\pi)=\frac{1}{2L}\max_k\left(\max_{j\le k} q_{\pi(j)}-q_{\pi(k)}\right). \tag{S2}$$

This is the smallest worst displacement divided by L for a nondecreasing point placement in the specified order, ignoring widths and gaps. It is an order obstruction, not the measured displacement of the final anchors.

For $i\ne j$, the displayed merge is the maximum profile value on the interval between x_i and x_j. Set $V^{\psi}_{ii}=\psi(x_i)$ and define

$$d_{\mathrm{top}}(\psi,T)=\max_{i,j}|V^{\psi}_{ij}-V_{ij}|, \qquad \mathrm{NMLE}=d_{\mathrm{top}}/\Delta_F, \tag{S3}$$

where $\Delta_F>0$ is the declared scalar range, shared across the sequence. We require

$$d_{\mathrm{top}}(\psi_t,T_t)\le\eta, \qquad \eta=(\varepsilon_0+\varepsilon)\Delta_F. \tag{S4}$$

Here ε_0 accounts for data resolution and $\varepsilon\ge 0$ is the additional analyst-selected budget. Exact topology below means zero labelled error on the declared tree; a numerical-tie allowance is accounted for through η, rather than assumed to be exactly zero on the original matrix.

If source and displayed trees are interpreted in the canonical cophenetic merge-tree representation, with fixed labels covering all their leaves and induced matrices including the diagonal values, the unnormalized discrepancy in Eq. S3 is the labelled interleaving distance [5, 2]. Under those representation conditions, $d_B\le d_I\le d_{\mathrm{top}}$ for the corresponding persistence diagrams and unlabelled trees [2, 4]. NMLE itself is a normalized discrepancy. If rendering introduces unlabelled extrema, the anchor matrix alone does not justify this identification for the entire displayed tree.

S3 Exact-Topology Bound

In a reduced tree whose proper ancestor levels strictly increase, zero labelled merge error requires every subtree's leaves to form a contiguous block. Otherwise a foreign leaf between two descendants would force a merge above that subtree's level. Equal-level events are treated as one node before imposing this requirement. For subtree X, let $m_X=\min_{i\in\mathscr{D}(X)} q_i$ and $M_X=\max_{i\in\mathscr{D}(X)} q_i$.

Theorem S1 (Exact-topology order bound)**.** *Every zero-error profile satisfies* $\mathrm{NROE}(\pi)\ge\nu^*(T,q)$*, where*

$$\nu^*(T,q)=\frac{1}{2L}\max_v\max_{\substack{X,Y\in\mathrm{ch}(v)\\ X\ne Y}}[\min(M_X-m_Y,M_Y-m_X)]^+. \tag{S5}$$

Recursively ordering each node's children by reference-hull midpoint $(m_X+M_X)/2$ *attains this bound for the point-order objective, in* $O(n\log n)$ *time.*

Proof sketch. If block X precedes block Y, an inversion from M_X to m_Y requires worst point displacement at least $[M_X-m_Y]^+/2$. Reversing the blocks gives $[M_Y-m_X]^+/2$. Every legal order must choose one orientation. Midpoint order chooses the less costly one since $m_X+M_X\le m_Y+M_Y$ is equivalent to $M_X-m_Y\le M_Y-m_X$. Every inverted leaf pair belongs to different child blocks at its lowest common ancestor, so the largest of these sibling costs also bounds all inversions from above in the midpoint order. □

The certificate depends only on the source tree and references, not on an observed layout or optimizer. It applies to any exact profile, including outputs of TMTM and ST-MTM with the same feature tree. Attainment refers to Eq. S2: it does not assert that a finite-width layout or its final quadratic placement attains $L\nu^*$.

S4 Width-Aware Bounds and Fixed-Order Evaluation

For nonoverlapping finite intervals, the fixed-order minimax cost is

$$\tau(\pi)=\min_{(x,z)\in\mathscr{L}(\pi)}\|x-q\|_\infty. \tag{S6}$$

The canvas is compact, so the minimum is attained when the layout set is nonempty. Widths, gaps, and canvas ends may force displacement even when the reference order is feasible in the point model. Set

$$\ell_i=q_i-W_i/2-H_i, \quad h_i=q_i+W_i/2+H_i, \quad s_i=W_i+g. \tag{S7}$$

For a nonempty leaf set S, define packing summaries

$$\begin{aligned} A(S)&=\max_{\alpha\in\{\ell_i:i\in S\}}\left(\alpha+\sum_{\substack{i\in S\\ \ell_i\ge\alpha}} s_i\right),\\ B(S)&=\max_{\zeta\in\{h_i:i\in S\}}\left(\sum_{\substack{i\in S\\ h_i\le\zeta}} s_i-\zeta\right). \end{aligned} \tag{S8}$$

Here $A(X)$ abbreviates $A(\mathscr{D}(X))$, and likewise for B.

Theorem S2 (Width-aware bound)**.** *If all leaves of X precede all leaves of Y, every valid placement has worst displacement at least $(A(X)+B(Y)-g)/2$. Consequently every exact-topology interval layout satisfies $\tau(\pi) \geq \mathscr{B}_w(T,q,W)$, where*

$$\mathscr{B}_w = \max\{0, B(\mathscr{F})-g, A(\mathscr{F})-g-L, K_w\},$$
$$K_w = \max_v \max_{X\neq Y\in \mathrm{ch}(v)} \frac{\min\{A(X)+B(Y), A(Y)+B(X)\}-g}{2}. \quad \text{(S9)}$$

$$z_b - z_a \geq \sum_{k\in S} s_k - g - W_a/2 - W_b/2. \quad \text{(S10)}$$

Here a,b enclose every selected interval in S; other intervals only increase the required span.

Proof. (i) Take $S_X \subseteq X$, $S_Y \subseteq Y$, let a be the first leaf of S_X and b the last leaf of S_Y in the order. All of $S = S_X \cup S_Y$ lies between them, so by (S10) and the anchor slack, $x_b - x_a \geq \sum_S s_k - g - W_a/2 - W_b/2 - H_a - H_b$. Since $\tau \geq q_a - x_a$ and $\tau \geq x_b - q_b$,

$$\tau \geq \tfrac{1}{2}\left[(q_a - x_a) + (x_b - q_b)\right] \geq \tfrac{1}{2}\left[\ell_a - h_b + \sum_{k\in S} s_k - g\right]$$
$$\geq \tfrac{1}{2}\left[\min_{S_X} \ell + \sum_{S_X} s_k + \sum_{S_Y} s_k - \max_{S_Y} h - g\right].$$

For $S_X = \{k \in X : \ell_k \geq \alpha\}$ the first two terms are at least $\alpha + \sum_{k\in X, \ell_k \geq \alpha} s_k$; for $S_Y = \{k \in Y : h_k \leq \zeta\}$ the last ones are at least $\sum_{k\in Y, h_k \leq \zeta} s_k - \zeta$. The two choices are independent, so maximizing each gives $A(X)$ and $B(Y)$. Each maximum is over upper (resp. lower) sets of one sorted list and is computed by one sort and a prefix sum. (ii) Let b be the last leaf of a set S. The intervals of S lie in $[0, z_b + W_b/2]$, so $z_b \geq \sum_S s_k - g - W_b/2$, $x_b \geq z_b - H_b$ and $\tau \geq x_b - q_b \geq \sum_S s_k - g - h_b \geq \sum_S s_k - g - \max_S h$. Lower sets of h give $B(\mathrm{all}) - g$. The right end is symmetric with the first leaf of S and gives $A(\mathrm{all}) - g - L$. (iii) Then $\ell = h = q$, $s = 0$, $A(X) = M_X$, $B(Y) = -m_Y$, and $\min(A(X)+B(Y), A(Y)+B(X))/2 = \min(M_X - m_Y, M_Y - m_X)/2$, which is L times the term of v^*. □

The summaries are evaluated by sorting and cumulative sums. With zero widths and gaps, $A(X) = M_X$ and $B(Y) = -m_Y$, and the bound reduces to Lv^*. For finite widths, Eq. S9 is a universal lower bound, not a formula for the globally optimal order or a promise of attainment.

For a specified order, the implementation also evaluates τ exactly. Index quantities in order as $w_k = W_{\pi(k)}$, $Q_k = q_{\pi(k)}$, and $H_k = \rho w_k/2$, and define packed left and right distances

$$D_k = \sum_{j<k} w_j + w_k/2 + (k-1)g, \quad R_k = \sum_{j>k} w_j + w_k/2 + (n-k)g.$$

The implemented interval-cost expression is

$$\tau(\pi) = \max\Big\{0,\ \max_k (D_k - Q_k - H_k),\ \max_k (Q_k - H_k - L + R_k),\ \max_{j<k} \frac{Q_j - Q_k + D_k - D_j - H_j - H_k}{2}\Big\}. \quad \text{(S11)}$$

These are boundary and pairwise packing requirements for the fixed-order interval problem. This exact subproblem cost, also available through a minimax linear program, is distinct from the tree-wide lower bound in Theorem S2.

Why the fixed-order evaluation is exact. Let $P = \sum_k w_k + (n-1)g \leq L$ and $s_0 = L - P$. Write the centers as $z_{\pi(k)} = D_k + y_k$. The interval constraints are equivalent to $0 \leq y_1 \leq \cdots \leq y_n \leq s_0$. An anchor within H_k of its center and within distance t of Q_k exists exactly when

$$Q_k - D_k - H_k - t \leq y_k \leq Q_k - D_k + H_k + t.$$

After intersecting these bounds with $[0, s_0]$, a nondecreasing sequence exists if and only if every prefix maximum of the lower bounds does not exceed the corresponding upper bound. Expanding this condition gives precisely the left-boundary, right-boundary and pairwise terms in Eq. S11, together with $t \geq 0$. Conversely, choosing prefix maxima supplies feasible centers at the largest of those terms, and each anchor can be chosen in the intersection of its center-slack interval and $[Q_k - t, Q_k + t]$. Thus the expression is attained. This explains the implemented fixed-order evaluation; it does not optimize the leaf permutation.

S5 Chain-Aware Separation and Bounds

Allowing $\eta > 0$ relaxes some displayed merge relationships, but does not eliminate all separation constraints. We use η in scalar units throughout this analysis to avoid confusing it with the normalized budget ε.

Lemma S1 (Separation)**.** *Let $i, j \in \mathscr{D}(u)$ and let leaf k lie strictly between their anchors. If $d_{\mathrm{top}} \leq \eta$, then $V_{ik} \leq h(u) + 2\eta$.*

Proof. The interval from i to k is contained in that from i to j, so $V_{ik} - \eta \leq V_{ik}^{\Psi} \leq V_{ij}^{\Psi} \leq V_{ij} + \eta \leq h(u) + \eta$. □

Let $A_\eta(u)$ be the highest ancestor of u with level at most $h(u) + 2\eta$. This ancestor is determined by the total level difference from u, not by testing each parent gap independently.

Theorem S3 (Chain-aware separation and bounds)**.** *No leaf outside $\mathscr{D}(A_\eta(u))$ can occur between two leaves of u in a profile with $d_{\mathrm{top}} \leq \eta$. Further, if*

$$\mathscr{D}(w) \cap \mathscr{D}(A_\eta(u)) = \varnothing, \quad \mathscr{D}(u) \cap \mathscr{D}(A_\eta(w)) = \varnothing, \quad \text{(S12)}$$

then all leaves of u precede all leaves of w, or conversely. For all such separated pairs (X,Y), including $(u, \{k\})$ with $k \notin \mathscr{D}(A_\eta(u))$, valid lower bounds are

$$\mathrm{NROE}(\pi) \geq \frac{1}{2L} \max_{(X,Y)} [\min(M_X - m_Y, M_Y - m_X)]^+,$$
$$\tau(\pi) \geq \max_{(X,Y)} \frac{\min\{A(X)+B(Y), A(Y)+B(X)\}-g}{2}. \quad \text{(S13)}$$

The order-independent canvas terms and zero remain part of the width-aware bound.

Proof sketch. A leaf outside $A_\eta(u)$ merges with each descendant of u above $h(u) + 2\eta$, so Lemma S1 excludes it from the span of u. Under Eq. S12, neither leaf set can interleave the other. If one set occupied both sides of the other, the latter would lie between two of its leaves, also contradicting separation. The two possible block orientations give Eq. S13 by the same inversion and packing arguments as before. □

A coarser certificate flattens internal nodes whose parent gaps are at most 2η and applies the exact-topology bounds to the remaining hierarchy. The chain certificate includes the separated pairs used by that bound and also accounts for accumulated gaps along ancestor chains. These certificates express necessary conditions for every budget-feasible layout; they do not construct a feasible order by themselves.

The relaxation concerns merge levels in the displayed profile. The input field, its extracted extrema, and supplied feature identities remain the reference data. The budget is an upper allowance, while measured NMLE records the amount actually used. Increasing it can enlarge the feasible set but does not guarantee smaller displacement for every feature, nor does it establish that a heuristic search returns a better layout at every budget.

For clarity, $C_\eta(T,q)$ is the right-hand side of the first inequality in Eq. S13. Let $K_{w,\eta}$ be the right-hand side of its second inequality. The full width-aware chain certificate is

$$C_{w,\eta}(T,q,W) = \max\{0, K_{w,\eta}, B(\mathscr{F}) - g, A(\mathscr{F}) - g - L\}. \quad \text{(S14)}$$

Thus both chain certificates include all eligible node–leaf pairs and mutually excluded node–node pairs; the width-aware one also includes the canvas-end terms. Singleton sets use $A(\{k\}) = \ell_k + s_k$ and $B(\{k\}) = s_k - h_k$. The ancestor test measures the full height difference, not each parent gap independently. All budget calculations use absolute scalar units η.

S6 Optimal Merge Barriers

Changing the leaf order changes which feature pairs span each inter-feature gap. Simply retaining the original LCA value of each adjacent pair need not minimize the resulting merge error. Let b_k be the barrier between leaves $\pi(k)$ and $\pi(k+1)$. The merge level displayed between positions $i < j$ is

$$V^{\Psi}_{\pi(i)\pi(j)} = \max_{i \le k < j} b_k. \quad \text{(S15)}$$

For fixed extremum values $v_i = V_{ii}$, define

$$C_k = \min_{i \le k < j} V_{\pi(i)\pi(j)}, \quad F_k = \max(v_{\pi(k)}, v_{\pi(k+1)}). \quad \text{(S16)}$$

Proposition S1 (Optimal barriers)**.** *The minimum labelled merge error for the fixed order and extremum values in the continuous non-strict model is*

$$\delta^*(\pi) = \max\left\{0,\ \max_k (F_k - C_k),\ \frac{1}{2}\max_{i<j}\left(V_{\pi(i)\pi(j)} - \max_{i \le k < j} C_k\right)\right\}, \quad \text{(S17)}$$

and $b_k = C_k + \delta^*(\pi)$ attains it.

Proof. The displayed merge of the ith and jth leaf is $\max_{i \le k < j} b_k$, and each barrier must be at least F_k to keep both basins. An error δ requires $b_k \le V_{\pi(i)\pi(j)} + \delta$ for every pair spanning k, i.e., $b_k \le C_k + \delta$; it also requires $\max_{i \le k < j} b_k \ge V_{\pi(i)\pi(j)} - \delta$ and $b_k \ge F_k$. The largest admissible barriers $b_k = C_k + \delta$ satisfy all constraints exactly when δ is at least each term of δ^*; smaller barriers only lower the maxima. Hence δ^* is optimal and attained. □

For fewer than two leaves the barrier list is empty and $\delta^* = 0$. For an exact-tree order, $\delta^* = 0$ and the barriers agree with the relevant adjacent LCA levels. For a conflicting order, the minimax solution balances raised and lowered displayed merges. This exact calculation provides the topology oracle in Eq. S4; it does not establish global optimality of the order search. The implemented direct evaluation costs $O(n^3)$ per order. The proposition permits equality at a leaf value. A raster implementation requiring strict extrema may add a small clearance, which must be counted separately in the measured rendered error.

Algorithm S1: Optimal Merge Barriers for a Fixed Leaf Order

Data: Order π and source merge matrix V, including original extremum values V_{ii}.
Result: Minimum continuous labelled merge error $\delta^*(\pi)$ and optimal barriers b.

```
Function OptimalBarriers(π, V):
    n ← |π|; δ ← 0
    if n < 2 then
        return (0, ∅)
    for k ← 1 to n − 1 do
        C_k ← min_{1≤i≤k<j≤n} V_{π(i)π(j)}
        F_k ← max{V_{π(k)π(k)}, V_{π(k+1)π(k+1)}}
        δ ← max{δ, F_k − C_k}
    for each pair 1 ≤ i < j ≤ n do
        δ ← max{δ, ½(V_{π(i)π(j)} − max_{i≤k<j} C_k)}
    for k ← 1 to n − 1 do
        b_k ← C_k + δ
    return (δ, b)
```

S7 Candidate Generation and Width-Aware Refinement

The following details specify the AnchorMTM heuristic over a restricted family of candidate orders. The source matrix is used by every merge-budget check. Midpoint initialization, threshold and conflict-driven candidates, and the retained point order are the same as in the main-paper algorithm.

Candidate structures. The helper returns the exact tree, threshold flattenings for distinct internal parent gaps up to 2η, and the retained conflict-driven selective flattening. Selective generation starts from the exact tree and considers releasing one or both internal blocks of the sibling pair with the largest order obstruction. Trials respect accumulated ancestor gaps, pass Eq. S4, and are compared by NROE, merge cost, and number of newly released nodes. Non-worsening NROE steps may continue the search; the best strictly improving state is retained. The point candidate π_p is the best feasible threshold midpoint order by NROE, replaced by the selective order only if it improves NROE. This is a finite heuristic candidate family, not enumeration of every permutation.

Refinement and selection. In the refinement, σ is the sum of squared positive boundary and pair terms in Eq. S11. Initial structures are ranked by $(\tau, \sigma, \text{number released})$. A refinement uses every insertion position for nodes with at most eight children and positions within distance three otherwise. A move must improve (τ, σ) lexicographically and pass the exact topology check; comparisons use the implementation's numerical tolerances. Retaining π_p ensures that the final (τ, NROE) selection cannot worsen its fixed-order displacement. Exact midpoint optimality, exact evaluation of τ, and heuristic leaf-order search are separate guarantees. Exhaustive enumeration certifies a global order optimum only when all permutations are covered.

The algorithm assumes the widths fit the canvas under Eq. S1. The exact-tree candidate supplies a feasible continuous topology order; width-capacity failure and insufficient raster resolution are reported by the corresponding routines. For $\eta = 0$, the certificate helper uses the exact-topology bounds; for $\eta > 0$, it evaluates the coarse and chain-aware certificates and retains the stronger applicable bound. These values are reported as benchmarks of unavoidable error. Candidate refinement and final selection use exact fixed-order costs; the certificates are not used as search-pruning thresholds.

S8 Complete Placement Model

After choosing π_t, we optimize anchors and interval centers over $\mathscr{L}(\pi_t)$. Let $P_t = \{(i,j) : i \text{ precedes } j \text{ in } \pi_t\}$, and let $\gamma'_i \in [0,1]$ be an optional confidence for a supplied temporal match. Default confi-

Algorithm S2: Width-Aware Refinement of One Candidate Structure

Data: Initialized structure S and order π; references, widths, source merge matrix and scalar budget.
Result: A budget-feasible refined order.
Function `RefineOrder`(S,π,q,W,V,η)**:**
  for *at most eight passes* **do**
    for *each internal node of S in traversal order* **do**
      Enumerate single-child-block insertions
      Find the first improving, budget-feasible move in (τ,σ) order
      Apply the move if one exists
    if *no move was accepted* **then**
      break
  return π

dence is one. The reference-aligned target is

$$y_i^t = x_{\mu_t(i)}^{t-1} + \Delta q_i^t, \qquad i \in \mathscr{C}_t.$$

The implemented normalized objective has four terms:

$$\begin{aligned} E_R &= \frac{1}{n_t}\sum_i (x_i^t - q_i^t)^2, \\ E_G &= \frac{1}{\max(1,|P_t|)} \sum_{(i,j)\in P_t} (x_j^t - x_i^t - d_{ij}^t)^2, \\ E_M &= \frac{1}{\max(1,|\mathscr{C}_t|)} \sum_{i\in\mathscr{C}_t} \gamma_i^t (x_i^t - y_i^t)^2, \\ E_C &= \frac{1}{n_t}\sum_i (x_i^t - z_i^t)^2. \end{aligned} \tag{S18}$$

Fixed order turns signed separations into linear expressions, so this is a convex quadratic objective. The cap enforces position fidelity while geometry and temporal terms choose among acceptable placements:

$$\begin{aligned} \min_{x_t,z_t} \quad & \lambda_R E_R + \lambda_G E_G + \lambda_M E_M + \lambda_C E_C \\ \text{subject to} \quad & (x_t,z_t) \in \mathscr{L}(\pi_t), \\ & \|x_t - q_t\|_\infty \le \tau(\pi_t) + \beta, \qquad \beta \ge 0. \end{aligned} \tag{S19}$$

In the formulas below, β denotes the fixed β_0. The parameter-class defaults are $(\lambda_R,\lambda_G,\lambda_M,\lambda_C) = (4,1,0.5,0.1)$, $\rho = 0.5$, and $\beta = 1$ in canvas units. Canvas extent, gap, and width scale are input-dependent quantities in those same units; their class defaults are 120, 0.5, and 0.06, respectively. They are not universal physical settings, and an evaluation must report any overrides. Equation S19 uses term means, as enabled by the default normalization flag, rather than unnormalized sums with the same numerical weights.

The reference-motion term penalizes $(x_i^t - x_{\mu_t(i)}^{t-1}) - \Delta q_i^t$. Equivalently, it favors continuity of the residual $x-q$, rather than a stationary displayed location when a feature moves. The code also supports stationary targets and intermediate motion fractions; Eq. S18 describes its default residual mode. Missing matches contribute no temporal penalty. Frames are processed sequentially using the previous solved anchors and references; this is not a joint global optimum over the entire sequence.

Since $\tau(\pi_t)$ is attained, the cap is feasible whenever the interval model is feasible. It limits the final worst displacement to $\tau(\pi_t) + \beta$ even when relative-distance objectives would favor a different absolute placement. It does not force each anchor to its reference or preserve every projected motion exactly. The solver uses linear feasibility/minimax subproblems and a constrained quadratic solve with numerical residual checks. Its fixed-order convexity does not extend a global guarantee to the discrete candidate search. A caller-selected absolute reference budget is a separate supported policy; Eq. S19 specifies the certificate-plus-slack construction used here.

S9 Profile Reconstruction and Verification

We reuse the augmented-tree sampling and scalar/path filling of the existing merge tree map pipeline [3, 1]. First, continuous intervals and anchors are mapped to a fixed-resolution canvas by one common affine conversion. Leaf extrema are placed at their anchors with their original scalar values. Regular samples of each leaf arc are resampled to the available interval positions and assigned alternately to either side of the anchor. If an arc has no regular samples, the existing renderer interpolates between its critical endpoint values. Thus leaf filling draws on original scalar data, but does not promise a bijective or lossless encoding of every source sample.

For consecutive leaf intervals, the original LCA determines the internal paths whose samples fill the intervening gap. The inherited renderer allocates positions along those paths and inserts the LCA level; direct sibling, direct-to-deep, and deep-to-deep cases differ in how many nonleaf paths contribute. This mechanism preserves the scalar hierarchy for a compatible exact order. In a budgeted order, however, original LCA filling alone does not guarantee the optimal error of Proposition S1.

The repository's optimal-filling path therefore corrects the rendered inter-anchor segments in join convention. For segment k, it clips interior values to the chosen barrier b_k and, if necessary, raises an existing peak to b_k. The anchors and their values remain fixed. This realizes Eq. S15 in the non-strict model; split-tree profiles are sign-restored afterward. Modified interior samples can differ from original scalar samples, and the minimax barrier formula does not impose an additional constraint keeping every barrier within the original scalar range. These are displayed scalar/topology changes, not changes to the input field. The implemented strict-extremum clearance may slightly raise a barrier above its non-strict optimum. Accordingly, δ^* is the exact continuous subproblem optimum, and the final raster must be measured separately.

All frames use the same reference direction, origin, extent, and pixel conversion. Stacking the resulting profiles gives

$$I(t,u) = \psi_t(u), \qquad 0 \le u \le L. \tag{S20}$$

No per-frame recentering or rescaling is applied to the displayed spatial axis. Raster rounding is checked against interval separation and available pixels; insufficient resolution is reported rather than repaired by moving the continuous layout. Temporal identity enters placement through the supplied tracks, while the stacked scalar values provide the temporal overview.

For each frame, reporting distinguishes the order certificate, a width-aware lower bound, the attained fixed-order cost $\tau(\pi_t)$, the final anchor displacement, $\delta^*(\pi_t)$, and the merge error measured from the rendered profile. The requested ε, numerical allowance ε_0, and placement slack β accompany these quantities. Rendered barriers and anchor rounding must be included when checking final budget compliance; a continuous certificate alone is not a raster certificate. Changed barriers can be identified by their discrepancy from the original adjacent merge level. Together these records expose how much of the displayed location is aligned with the reference and how much displayed merge structure has changed, without treating the budget itself as an observed error.

Continuous and rendered acceptance. Exact Mode sets $\varepsilon = 0$ and keeps only the numerical allowance $\eta = \varepsilon_0 \Delta_F$, which merges saddles tied to within $\varepsilon_0 \Delta_F$; strict exact topology is $\eta = 0$. The exact-tree structure is retained among refinement seeds, but capacity and numerical failures can still stop layout. The renderer reports

invalid raster capacity and nonfinite values. Optimal filling may introduce strict-extremum clearance and records measured budget discrepancies and topology diagnostics. There is no uniform final budget/label-coverage rejection or repair gate. Applying the full labelled-tree interpretation requires checking that labels cover the rendered tree; a continuous barrier optimum alone does not establish this.

S10 Per-Frame Distributions over the Merge Budget

The tables of the paper report means over frames; Figs. S1–S4 show the frames behind them. The planar runs cover the complete sequences with the references, widths, gaps, weights and raster lengths of the paper's four-dataset comparison table, and only ε changes. The run at $\varepsilon = 0$ is Exact Mode and the run at the selected budget (Ring 1%, Wildfires 5%, Storms 10%) is Balanced Mode; both coincide value by value with the table runs. For frame t, the worst displacement is $\max_i |x_i^t - q_i^t|/L_{world}$ on rendered anchors, and its mean over frames is the table's NPE; the merge error is the attained fixed-leaf merge error, including diagonal extrema, divided by the scalar range; SNS is the frame's scale-normalized stress. Every run passes its continuous and rendered budget checks. Gaussian gives the same map at every budget and is omitted.

Figure S1 extends the sweep of the paper to 30% (Ring 100%) and adds SNS. On Storms the upper tail keeps shrinking up to 15%, where the worst frame reaches 4.08%, and nothing changes beyond; 294 of 744 frames then use the allowance. SNS is a by-product rather than a target of the budget: it falls on Ring (0.149 at $\varepsilon = 0$, 0.129 at 10%) and slightly on Storms (0.079 to 0.074 at 10%, 0.076 from 15%), but rises on Wildfires (0.043 to 0.045). On Ring the last frame keeps a displacement of 10.02% until $\varepsilon = 63\%$; enumerating all 120 orders of that five-leaf frame places its continuous release threshold at 62.0014%.

Figure S2 shows which frames change. The 294 Storms frames that use the allowance at some budget form 24 runs of consecutive hours, and the budget at which a frame first changes differs within a run. The two runs whose worst displacement still exceeds 9% at $\varepsilon = 10\%$, frames 151–177 and 305–345, are released at 15%. All other frames keep the Exact layout at every budget.

In the volumes almost every frame contains a conflict that a budget can release (Fig. S3): 200 of 201 Tangaroa frames at 2% and 5% and all of them from 10%, and 98 of 102 Cylinder frames at every positive budget. The whole distribution therefore moves. The Cylinder median falls from 6.76% to 1.04% already at 2%, while the Tangaroa median falls gradually from 11.32% to 1.20% at 20%, in line with its deep conflicts in Section S11.

Figure S4 shows the selection rule of the paper's Mode selection paragraph. On Wildfires 5% gives the smallest validation NPE at a small SNS cost and is the smallest budget within the tolerance; on Storms validation NPE falls at every step up to 10%, SNS changes little beyond 6%, and J is smallest at 10%. The first quarter of Ring contains no conflict that a budget releases (Fig. S2), so its selection keeps 1%, while later frames change at larger budgets.

S11 Bound Evaluation on an Earlier Benchmark

Section 5.5 of the paper evaluates the bounds on the frames of an earlier benchmark, prepared with the same extractor, matching and reference estimator and with one default configuration Θ_0 (Section S8) on every sequence. Table S1 lists the sequences and, for four budgets, the frames on which the interval cost τ of the selected order equals the lower bound; for $\varepsilon > 0$ the bound is the larger of the bounded-topology certificate and the flattening certificate (Section S5). The 12-hourly Storms sequence and the 30-frame Wildfires sequence are prepared differently from the hourly Storms and 32-frame Wildfires inputs of the main comparison, and Ring follows the TMTM generator.

Table S2 splits the order error of TMTM and ST-MTM on the same frames. Both keep the merge tree exactly, so Theorem S1 applies: midpoint ordering of the same tree attains ν^*, and the share of the mean NROE above the mean ν^* is chosen by the order rule rather than forced by the hierarchy. TMTM is a port of its official implementation; ST-MTM follows its published description with temporal reordering threshold 0.95 and meets the optimality conditions of its quadratic programs (relative KKT residuals below 10^{-6}). Neither method fixes the direction of its axis, so each map is evaluated in the global mirror that agrees better with the reference. Over all temporal reordering thresholds from 0 to 1, each in its favourable mirror, the ST-MTM share stays within 53–95% on the real planar sequences and within 37–61% on Ring; shares are computed from unrounded means.

Figure S5 places every compared map of the earlier benchmark in one plane. The bound is evaluated on 126 allowances and the AnchorMTM order on 43, both without numerical allowance, so at $\varepsilon = 0$ the bound is the strict ν^*; at the nine allowances of Table S1 the computation reproduces the saved results exactly. At $\varepsilon = 0$ the AnchorMTM order attains ν^* on all 917 frames. The shape of the edge follows Theorem 3: where conflicting subtrees merge just below their parent, a small allowance releases them (Wildfires: 5.80% to 2.77% at 1%; Cylinder: 12.24% to 0.23% at 1%, and to 6.15% already below 10^{-6} of the scalar range, the precision of the stored data), while deep conflicts stay expensive (Tangaroa above 7% up to 5%; Storms 1.43% at 10%).

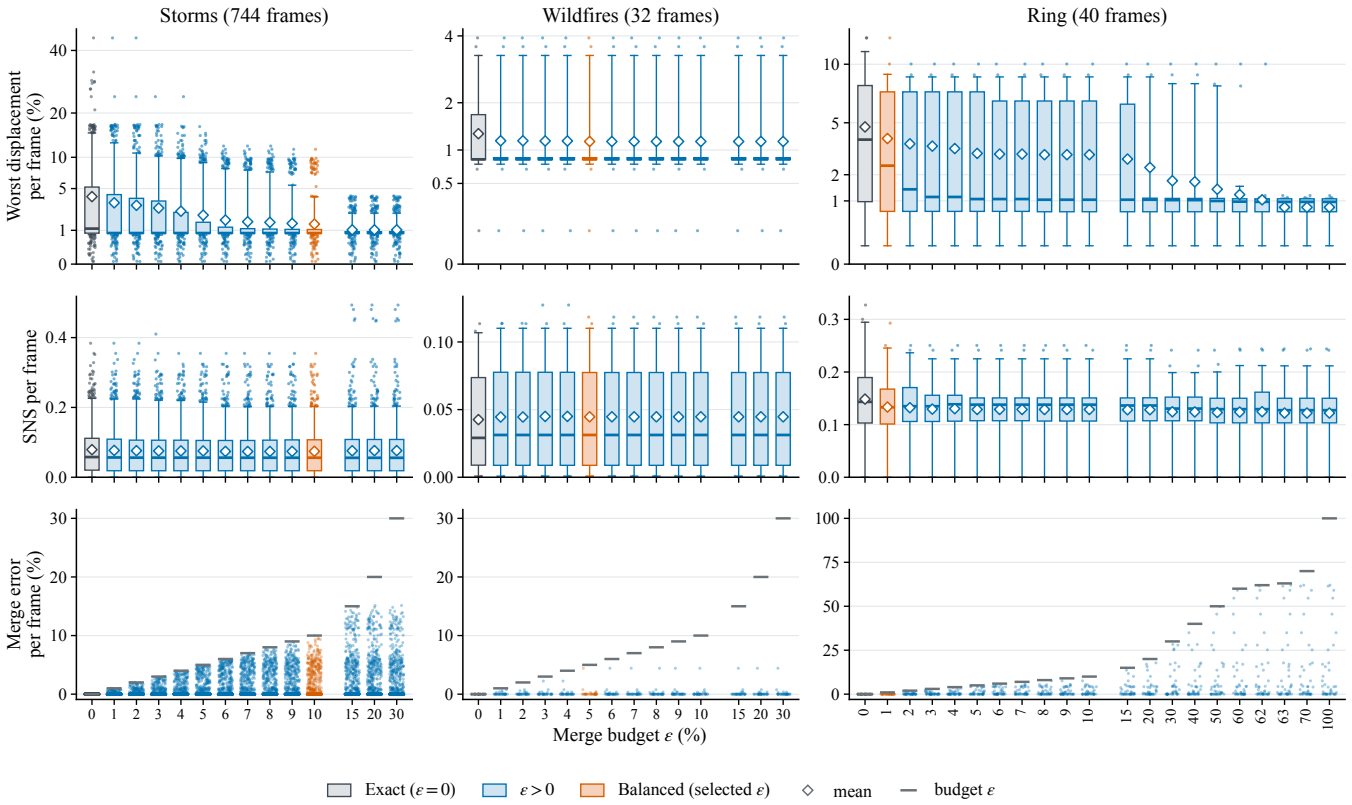


Figure S1: Per-frame distributions over the complete budget sweep. Rows: worst displacement per frame (square-root scale), SNS per frame (frames with too few features omitted), and attained merge error per frame with its budget (grey dash). Budgets above 10% use larger steps (gap in the axis); Ring is swept to 100% to locate its last release. Boxes: quartiles; whiskers: 5th–95th percentiles; dots: other frames; diamonds: means.

Table S1: Earlier benchmark and bound attainment: frames on which τ of the selected order equals the lower bound. Leaves: retained extrema per frame.

Sequence	Scalar field	Frames	Leaves	Tree	$\varepsilon = 0$	2%	5%	10%
Storms	ERA5 sea-level pressure, 2D	118	1–8	join	116	116	110	110
Wildfires	$\log(1 + \mathrm{FRP})$, 2D	30	6–11	split	25	27	24	24
OLR	outgoing longwave radiation, 2D	60	2–8	join	60	60	60	59
$PM_{2.5}$	$PM_{2.5}$ concentration, 2D	366	1–7	split	365	361	363	363
Ring	synthetic ring intensity, 2D	40	1–7	split	40	40	40	40
Planar total		614			606	604	597	596
Tangaroa	velocity magnitude, 3D	201	18–93	join	27	24	30	24
Cylinder	Q criterion, 3D	102	4–75	split	81	77	76	74

Table S2: Order error of exact-topology maps on the earlier benchmark (percent of the canvas, means over frames) and the share above the strict bound ν^*.

Sequence	ν^*	TMTM NROE	share	ST-MTM NROE	share
Storms	3.77	18.31	79	16.28	77
Wildfires	5.80	19.65	70	18.72	69
OLR	0.89	19.31	95	16.59	95
$PM_{2.5}$	0.97	4.44	78	5.21	81
Ring	2.66	7.18	63	4.47	41
Tangaroa	9.24	19.50	53	15.80	41
Cylinder	12.24	26.62	54	17.77	31

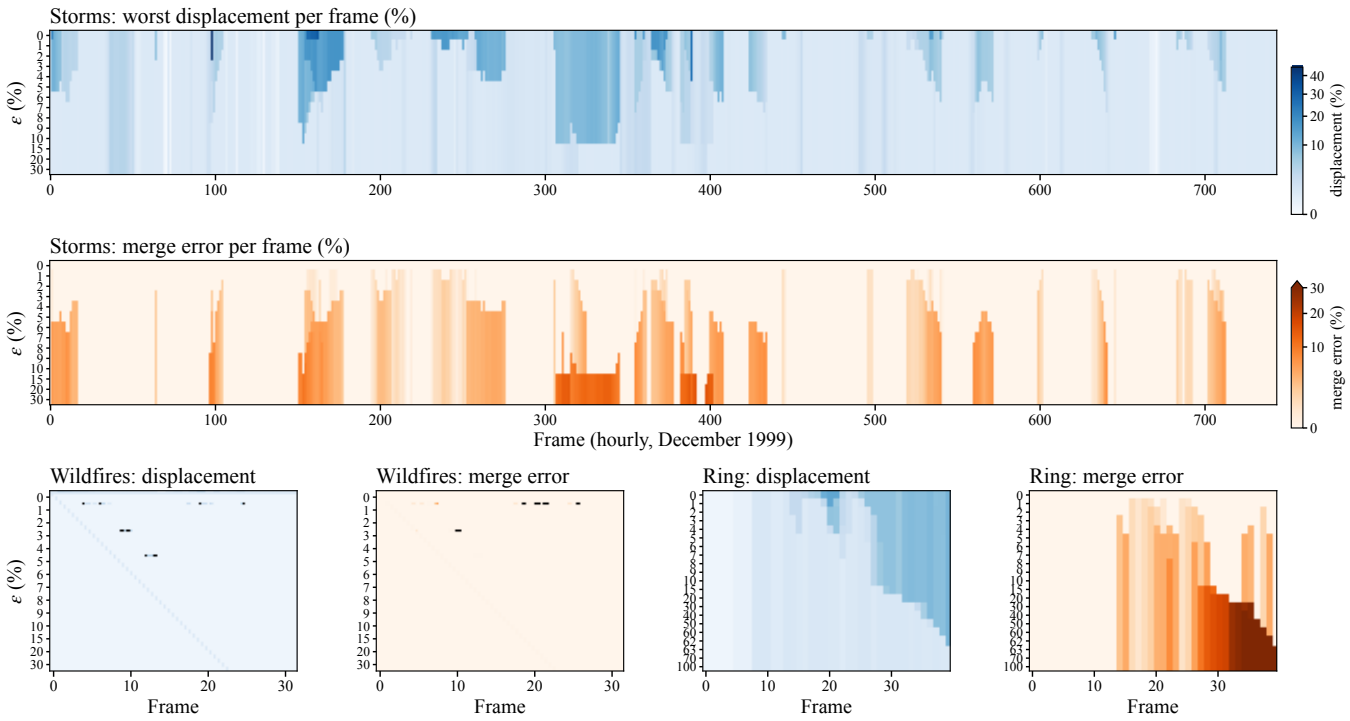


Figure S2: Which frames change. Each row is one budget and each column one frame; colour gives the worst displacement (blue) or the attained merge error (orange) of that frame. Square-root colour scales are shared by all panels; merge errors above 30% (Ring only) saturate. Storms frames are hourly from 1 to 31 December 1999.

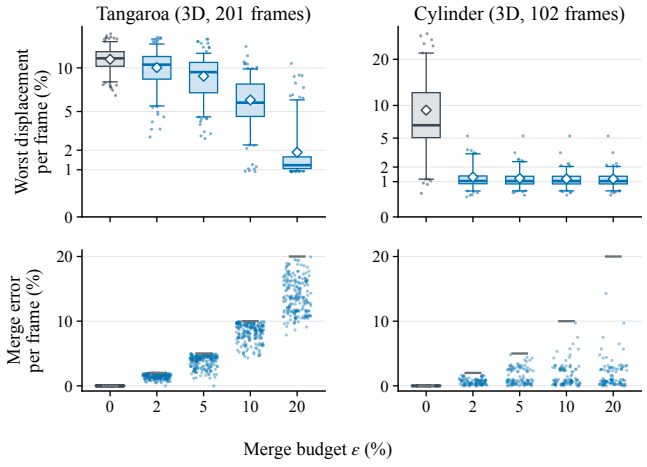


Figure S3: Volumetric budget sweeps with one default configuration: weights $(4, 1, 0.5, 0.1)$, $\rho = 0.5$, 8192-position rasters, and references from matched-extremum motion (Tangaroa) or the streamwise x axis (Cylinder). These settings differ from the tuned configurations of the paper's volumetric table. All 1,515 rendered frames were rechecked against the fixed-leaf merge matrices; none exceeds its budget. Encodings as in Fig. S1.

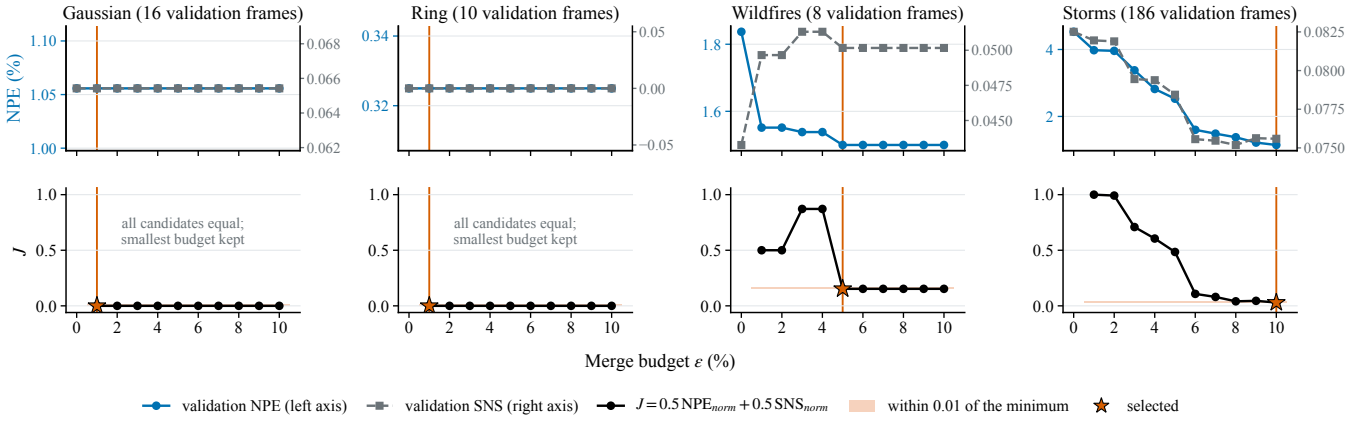


Figure S4: Selection of the Balanced budget on the first chronological quarter of each sequence (validation frames). Top: validation NPE (left axis) and SNS (right axis) for $\varepsilon = 0$–10%. Bottom: the score J of the paper over the eligible positive budgets, the band within 0.01 of its minimum, and the selected budget (star), frozen before the full runs. On the Gaussian and Ring validation frames every candidate gives the same map, so the smallest budget is kept.

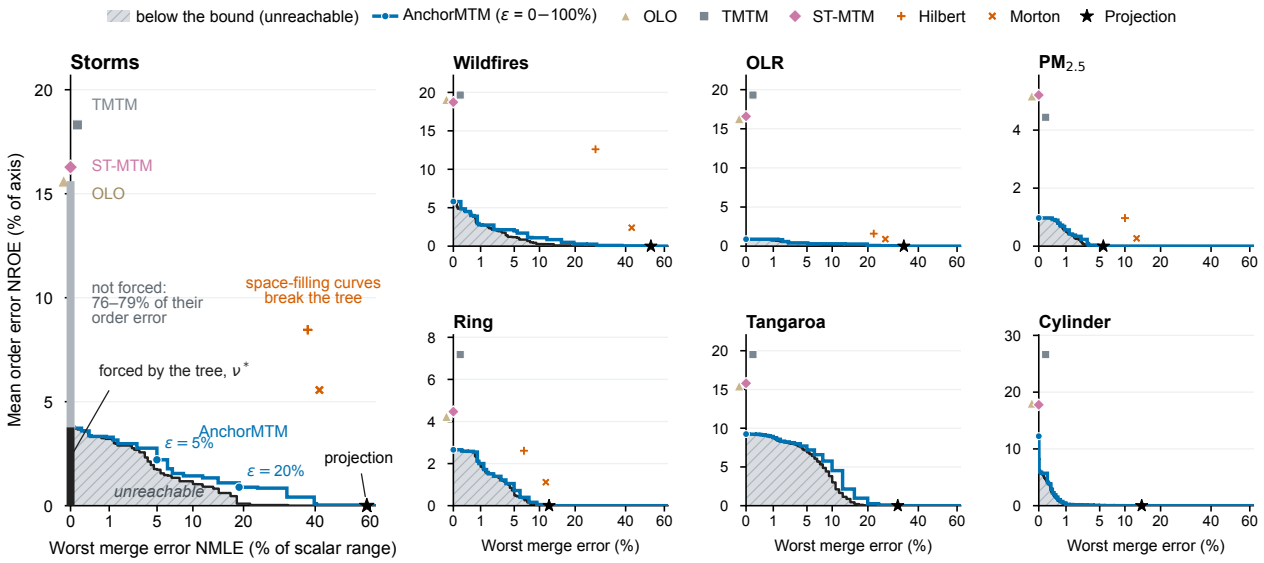


Figure S5: The position–topology plane: no map can enter the grey region, topology-preserving maps lie far above it, and AnchorMTM runs along its edge. Every map of a sequence is one point: its worst merge error over all frames (horizontal, square-root scale) and its mean order error NROE (vertical). Grey: below the frame-averaged bound (the larger of Theorem 3 of the paper and the flattened bound, with Theorem 1 at zero merge error); no map with that worst merge error has a smaller mean order error. Blue: AnchorMTM orders at allowances ε from 0 to 100%, drawn as the least order error reached at each merge error. TMTM, ST-MTM and optimal leaf ordering (OLO) keep the tree and lie at zero merge error (offset slightly for legibility), each in its better mirror; Hilbert and Morton curves (planar sequences only) and the projection follow space and break the tree. Storms panel: the black part of the axis is the forced ν^*, the grey part the share of the order error that the three order rules choose. On Storms, OLR, $PM_{2.5}$ and Ring every frame has at most eight leaves; enumerating all leaf orders shows that the blue line is the exact optimum at every tested allowance, except one Storms frame at $\varepsilon = 31.6\%$ (1.9 percentage points), so there the band between it and the grey region lies in the bound.

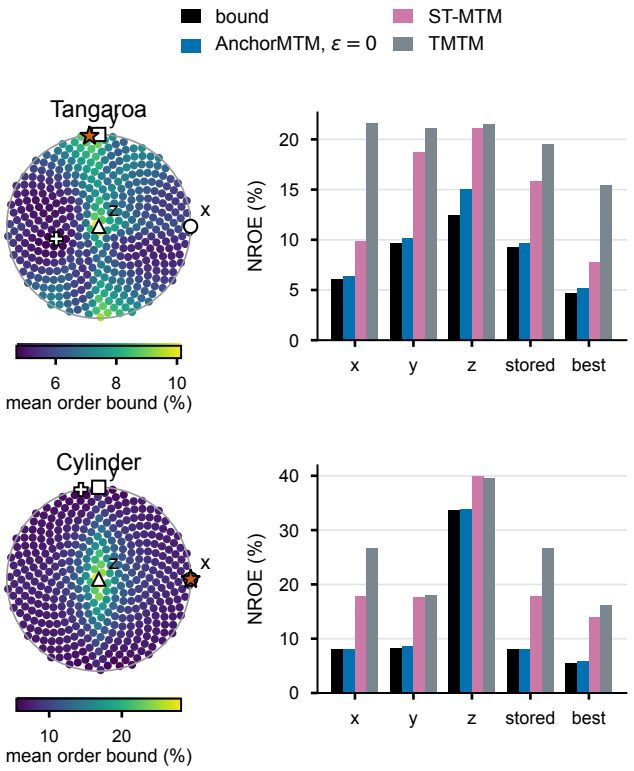


Figure S6: The forced cost depends on the reference direction and can be computed before a map is drawn. Left: mean order bound at the numerical allowance ε_0 over 400 directions of the hemisphere (equal-area projection; centre z, rim the xy plane); markers: coordinate axes, the stored direction (star) and the smallest bound (cross). Right: the bound and the mean order error of AnchorMTM at $\varepsilon = 0$, ST-MTM and TMTM in these directions. The bound spans 4.70–10.14% on Tangaroa and 5.49–28.15% on Cylinder.

S12 Exploratory Volumetric Case: Central-Tree Guidance

The central-tree example is a separate qualitative configuration, excluded from both quantitative tables of the paper. Cylinder comprises 102 volumes ($135 \times 64 \times 48$) of the Q-criterion with 2% per-frame persistence. It tests how additional tree guidance changes the map's coordinate interpretation.

The variant in Fig. S7 uses the same 102 volumes, retained 2% persistence trees and a 1% merge allowance. Full-sample tree coordinates $g_i^t \in [-0.5, 0.5]$ add $16\,\mathrm{mean}_i(x_i^t - g_i^t)^2$ to the placement objective; reference, geometry, temporal and eccentricity weights are 1, 1, 0.5 and 0.1. True-X targets retain one fixed domain transform. This changes multiple weights and is not a single-factor ablation.

Its fixed tree-compatible order needs a worst-case reference cap of 55.21%; we explicitly allow 56.21%, including 1% slack. Monotone warping of the tree profile through the optimized intervals produces 32768-position rasters. All retained-pair checks yield $\mathrm{NMLE}_{max} = 0$, within 1%, including half-pixel position rounding; this conservative order does not exploit budgeted reordering or introduce simulation frames.

Central guidance exposes unfolding but sacrifices anchoring. Frame-balanced mean absolute X error is 21.1%, 25.5% and 28.1% for guidance weights 1, 4 and 16, versus 1.9% for the Cartesian-reference configuration. This differs from NPE and pooled mean displacement. These outputs are not the Cartesian-reference configurations of the paper's volumetric table. Equal-X features may differ in Y, Z; their layout separation does not encode depth.

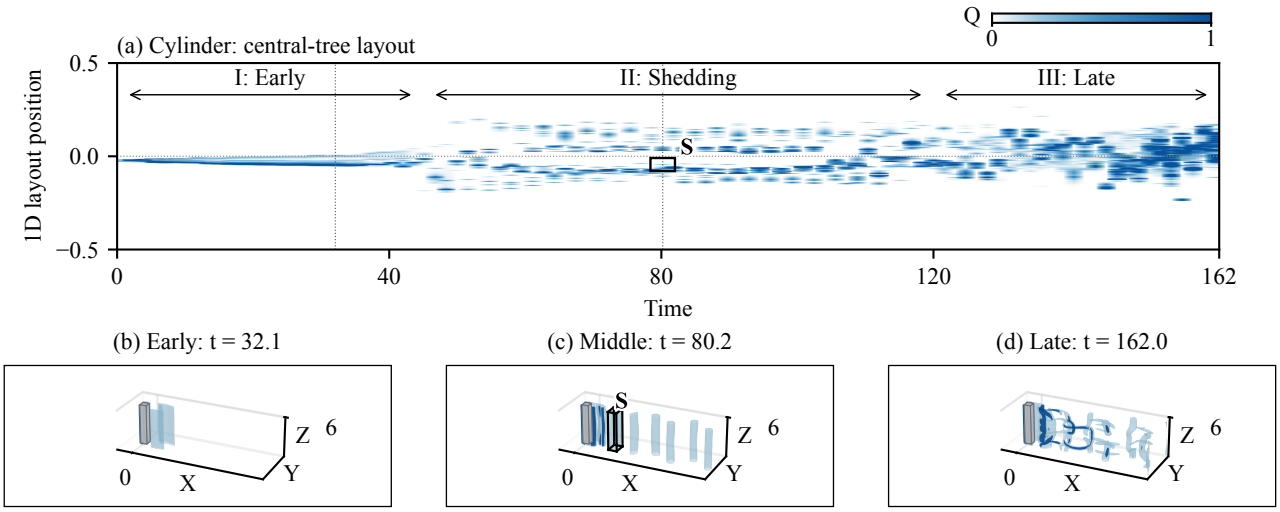


Figure S7: Cylinder central-tree variant: the full 102-frame map (a) and source Q isosurfaces at three times (b–d), with a common camera and blue scalar scale. $Q = 0.2$ is translucent and $Q = 1$ marks the core; values above 1 saturate. The outlined component contains extremum S. Map position is not physical X; the padded locator and stages I–III are viewing guides, not exact support projections or transition estimates.

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